HUMAN RESOURCES (HR) MANAGEMENT MODULE

SmartCampus School Management System
Technical Specification & Implementation Guide — v1.0 · March 2026
Confidential · Internal Documentation

Audience: Development Team · HR Department · School Administrators

Related Modules:
Staff Management · Payroll · Finance · Attendance · Super Admin · Security & Compliance

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1. MODULE OVERVIEW

1.1 Purpose

The **Human Resources (HR) Management Module** manages all employee lifecycle operations within the institution.

The module supports:

- Employee records
- Recruitment and onboarding
- Attendance monitoring
- Leave management
- Payroll coordination
- Performance evaluations
- Training and development
- HR analytics

The system ensures **efficient workforce administration, compliance with employment regulations, and strategic staff management**.

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2. CORE OBJECTIVES

The HR module enables institutions to:

1. Maintain **complete employee records**
2. Manage **staff recruitment and onboarding**
3. Track **attendance and leave**
4. Coordinate **payroll processes**
5. Conduct **performance evaluations**
6. Manage **training and development**
7. Provide **HR analytics and reporting**
8. Ensure **regulatory compliance**

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3. SYSTEM ROLES & ACCESS CONTROL

Role	Permissions
Super Admin	Platform-wide HR oversight
School Admin	Manage HR configurations
HR Manager	Full HR management
HR Officer	Operational HR tasks
Department Head	Manage department staff
Employee	View personal HR records
Auditor	Read-only HR access

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4. MODULE ARCHITECTURE

The HR Management Module consists of the following subsystems:

```
Human Resources Management
│
├── HR Dashboard
├── Employee Directory
├── Recruitment Management
├── Onboarding Management
├── Attendance Management
├── Leave Management
├── Payroll Coordination
├── Performance Management
├── Training & Development
├── HR Reports & Analytics
├── Compliance & Audit Logs
```

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5. HR DASHBOARD

5.1 Overview

The **HR Dashboard** provides a centralized view of employee data and HR operations.

Dashboard Metrics

- Total Employees
- Active Employees
- Employees on Leave
- New Hires
- Attendance Rate
- Pending Leave Requests

Example Dashboard

```
-----------------------------------------
HR Dashboard
-----------------------------------------

Total Employees: 150
Active Employees: 140
Employees on Leave: 10

Attendance Rate: 95%

Pending Leave Requests: 6
New Hires (This Month): 4
```

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6. EMPLOYEE DIRECTORY

Provides a searchable list of all employees.

Features

- Staff search
- Department filters
- Role filters
- Contact directory

Example Directory Entry

```
Name: Michael Adams
Position: English Teacher
Department: Academics
Email: michael.adams@school.edu
Phone: +254 XXX XXX
Status: Active
```

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7. EMPLOYEE PROFILE MANAGEMENT

Each employee has a detailed profile.

Profile Components

- Personal information
- Employment details
- Contact information
- Qualifications
- Certifications
- Employment documents

Data Model

```
Employee
- employee_id
- full_name
- department_id
- position
- employment_type
- hire_date
- contact_information
- qualifications
- status
```

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8. RECRUITMENT MANAGEMENT

Handles job openings and hiring processes.

Recruitment Workflow

```
Job Opening Created
      ↓
Applications Submitted
      ↓
Candidate Screening
      ↓
Interview Process
      ↓
Hiring Decision
      ↓
Employee Onboarding
```

Data Model

```
JobApplication
- application_id
- candidate_name
- position_applied
- resume
- application_status
```

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9. ONBOARDING MANAGEMENT

Manages the process of integrating new employees.

Onboarding Tasks

- Document submission
- Contract signing
- Account creation
- Training orientation

Data Model

```
OnboardingTask
- task_id
- employee_id
- task_description
- completion_status
```

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10. ATTENDANCE MANAGEMENT

Tracks daily staff attendance.

Attendance Methods

- Biometric devices
- RFID cards
- Mobile check-in
- Manual entry

Attendance Data Model

```
EmployeeAttendance
- attendance_id
- employee_id
- check_in_time
- check_out_time
- attendance_status
- date
```

Statuses include:

- Present
- Absent
- Late
- On Leave

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11. LEAVE MANAGEMENT

Allows employees to request leave.

Leave Types

- Annual leave
- Sick leave
- Maternity leave
- Study leave
- Emergency leave

Leave Workflow

```
Employee submits request
        ↓
Department Head Review
        ↓
HR Approval
        ↓
Leave Recorded
```

Data Model

```
LeaveRequest
- leave_id
- employee_id
- leave_type
- start_date
- end_date
- approval_status
```

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12. PAYROLL COORDINATION

Integrates HR data with payroll systems.

Payroll Components

- Basic salary
- Allowances
- Deductions
- Taxes

Payroll Data Model

```
PayrollRecord
- payroll_id
- employee_id
- salary
- allowances
- deductions
- pay_period
```

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13. PERFORMANCE MANAGEMENT

Tracks employee performance reviews.

Evaluation Criteria

- Teaching effectiveness
- Administrative efficiency
- Student feedback
- Attendance consistency

Data Model

```
PerformanceReview
- review_id
- employee_id
- review_period
- rating
- comments
```

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14. TRAINING & DEVELOPMENT

Manages employee skill development.

Training Types

- Professional development
- Teaching methodology
- IT systems training
- Leadership training

Data Model

```
TrainingProgram
- training_id
- program_name
- trainer
- training_date
```

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15. HR REPORTS & ANALYTICS

Standard Reports

- Employee distribution by department
- Attendance trends
- Leave utilization
- Recruitment statistics
- Training participation

Example Report

```
Attendance Report (April)

Total Employees: 150
Average Attendance: 96%

Departments with Highest Attendance:
Academics
Administration
```

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16. AUTOMATION FEATURES

Automation improves HR efficiency.

Examples include:

- Automated leave approvals
- Payroll calculation integration
- Attendance alerts
- Contract expiry notifications

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17. COMPLIANCE & SECURITY

Security Features

- Role-based access control
- Encrypted employee data
- Multi-factor authentication
- Secure document storage

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18. AUDIT TRAIL

All HR actions are logged.

Data Model

```
AuditLog
- log_id
- user_id
- action
- employee_id
- timestamp
```

Example entry:

```
User: HR Manager
Action: Updated Employee Record
Employee: Michael Adams
Date: 16 March 2026
```

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19. SYSTEM INTEGRATIONS

Module	Integration Purpose
Staff Management	Employee profiles
Payroll	Salary processing
Finance	Payroll accounting
Attendance	Staff check-in system
Super Admin	Multi-campus HR management

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20. SCALABILITY

The HR system supports:

- Large employee databases
- Multi-campus institutions
- Cloud-based deployments
- High-volume attendance tracking

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21. MOBILE HR SUPPORT

Mobile access allows employees to:

- Submit leave requests
- Check attendance records
- View HR notifications
- Access training materials

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22. DATA IMPORT & EXPORT

Supported formats:

- CSV
- Excel
- API integration

Use cases include:

- Bulk employee imports
- Payroll exports
- HR reporting

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23. FUTURE ENHANCEMENTS

Advanced capabilities include:

- AI workforce analytics
- Predictive staff retention analysis
- Automated recruitment screening
- Smart attendance anomaly detection
- HR chatbot support

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